Q1 2021 (01 Sep Shift 2)

Let $\mathbf{J}_{n,m}=\int_0^{\frac{1}{2}} \frac{x^n}{x^m-1} dx, \quad \forall n>m$ and $n, m \in N$.

Consider a matrix $A=[a_i]_{3\times 3}$ where

$a_{ij}=\begin{cases} J_{6+i,3}-J_{i+3,3}, & i\le j \\ 0 & i>j \end{cases}$. Then $|\operatorname{adj} A^{-1}|$ is :

(1) $(15)^2\times 2^{42}$

(2) $(15)^2\times 2^{34}$

(3) $(105)^2\times 2^{38}$

(4) $(105)^2\times 2^{36}$

Q2 2021 (31 Aug Shift 2)

The number of elements in the set

$\left\{A=\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in \{-1,0,1\} \text{ and } (I-A)^3=I-A^3\right\}$,

where I is 2×2 identity matrix, is :

Q3 2021 (27 Aug Shift 2)

Let $A=\begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, where[t]

denotes the greatest integer less than or equal to t. If $\det(A)=192$, then the set of values of x is the interval:

(1) $[68,69)$

(2) $[62,63)$

(3) $[65,66)$

(4) $[60,61)$

Q4 2021 (27 Aug Shift 1)

If the matrix $A=\begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ satisfies $A\left(A^3+3I\right)=2I$ then the value of K is :

(1) $\frac{1}{2}$

(2) $-\frac{1}{2}$

(3) -1

(4) 1

Q5 2021 (26 Aug Shift 2)

Let A be a 3×3 real matrix.

If $\det(2\operatorname{Adj}(2\operatorname{Adj}(\operatorname{Adj}(2\mathrm{A})))) = 2^{41}$, then the value of $\det\left(\mathrm{A}^2\right)$ equal .

Q6 2021 (26 Aug Shift 2)

Let $\mathrm{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $\mathrm{A}^{2025} - \mathrm{A}^{2020}$ is equal to :

(1) $A^6 - A$

(2) A^5

(3) $A^5 - A$

(4) A^6

Q7 2021 (26 Aug Shift 1)

If $\mathrm{A} = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $\mathrm{B} = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$, and

$\mathrm{Q} = \mathrm{A}^{\mathrm{T}}\mathrm{BA}$, then the inverse of the matrix $\mathrm{A}\, Q^{2021} A^{\mathrm{T}}$ is equal to :

(1) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

(2) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

(3) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(4) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

Answer Key

Q1 (3)	**Q2** (8)	**Q3** (2)	**Q4** (1)
Q5 (4)	**Q6** (1)	**Q7** (2)	

Q1 (3)

$$\begin{bmatrix} \surd & \surd & \surd \\ a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$J_{6+i,3} - J_{i+3,3}; i \leq j$

$\Rightarrow \int_0^{\frac{1}{2}} \frac{x^{6+i}}{x^3-1} - \int_0^{\frac{1}{2}} \frac{x^{i+3}}{x^3-1}$

$\Rightarrow \int_0^{1/2} \frac{x^{i+3}(x^3-1)}{x^3-1}$

$\Rightarrow \frac{x^{3+i+1}}{3+i+1} = \left(\frac{x^{4+i}}{4+i}\right)_0^{1/2}$

$a_{ij} = j_{6+i,3} - j_{i+3,3} = \frac{\left(\frac{1}{2}\right)^{4+i}}{4+i}$

$a_{11} = \frac{\left(\frac{1}{2}\right)^5}{5} = \frac{1}{5.2^5}$

$a_{12} = \frac{1}{5.2^5}$

$a_{13} = \frac{1}{5.2^5}$

$a_{22} = \frac{1}{6.2^6}$

$a_{23} = \frac{1}{6.2^6}$

$a_{33} = \frac{1}{7.2^7}$

$$A = \begin{bmatrix} \frac{1}{5.2^5} & \frac{1}{5.2^5} & \frac{1}{5.2^5} \\ 0 & \frac{1}{6.2^6} & \frac{1}{6.2^6} \\ 0 & 0 & \frac{1}{7.2^7} \end{bmatrix}$$

$|\mathrm{A}| = \frac{1}{5.2^5}\left[\frac{1}{6.2^6} \times \frac{1}{7.2^7}\right]$

$|\mathrm{A}| = \frac{1}{210.2^{18}}$

$\left|\mathrm{adj} \mathrm{A}^{-1}\right| = \left|\mathrm{A}^{-1}\right|^{\mathrm{n}-1} = \left|\mathrm{A}^{-1}\right|^2 = \frac{1}{(|\mathrm{A}|)^2}$

$\Rightarrow \left(210.2^{18}\right)^2$

$(105)^2 \times 2^{38}$

Q2 (8)

$(I-A)^3 = I^3 - A^3 - 3A(I-A) = I - A^3$

$\Rightarrow 3A(I-A) = 0$ or $A^2 = A$

$\Rightarrow \begin{bmatrix} a^2 & ab+bd \\ 0 & d^2 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$

$\Rightarrow a^2 = a, b(a+d-1) = 0, d^2 = d$

If $\mathrm{b} \neq 0, \mathrm{a}+\mathrm{d}=1 \Rightarrow 4$ ways

If $\mathrm{b}=0, \mathrm{a}=0,1 \&\ \mathrm{d}=0,1 \Rightarrow 4$ ways

$\Rightarrow$ Total 8 matrices

Q3 (2)

$\begin{vmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{vmatrix} = 192$

$\mathrm{R}_1 \rightarrow \mathrm{R}_1 - \mathrm{R}_3 \& \mathrm{R}_2 \rightarrow \mathrm{R}_2 - \mathrm{R}_3$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ [x] & [x]+2 & [x]+4 \end{bmatrix} = 192$

$2[x] + 6 + [x] = 192 \Rightarrow [x] = 62$

Q4 (1)

Given matrix $A = \begin{bmatrix} 0 & 2 \\ k & -1 \end{bmatrix}$

$\mathrm{A}^4 + 3\mathrm{IA} = 2\mathrm{I}$

$\Rightarrow \mathrm{A}^4 = 2\mathrm{I} - 3\ \mathrm{A}$

Also characteristic equation of A is

$|\mathrm{A} - \lambda \mathrm{I}| = 0$

$\Rightarrow \begin{vmatrix} 0-\lambda & 2 \\ \mathrm{k} & -1-\lambda \end{vmatrix} = 0$

$\Rightarrow \lambda + \lambda^2 - 2\mathrm{k} = 0$

$\Rightarrow \mathrm{A} + \mathrm{A}^2 = 2\ \mathrm{K}.\mathrm{I}$

$\Rightarrow \mathrm{A}^2 = 2\mathrm{KI} - \mathrm{A}$

$\Rightarrow \mathrm{A}^4 = 4\ \mathrm{K}^2\mathrm{I} + \mathrm{A}^2 - 4\mathrm{AK}$

Put $\mathrm{A}^2 = 2\mathrm{KI} - \mathrm{A}$

and $\mathrm{A}^4 = 2\mathrm{I} - 3\ \mathrm{A}$

$2I - 3A = 4K^2I + 2KI - A - 4AK$

$\Rightarrow I\left(2 - 2K - 4K^2\right) = A(2 - 4K)$

$\Rightarrow -2I\left(2K^2 + K - 1\right) = 2A(1 - 2K)$

$\Rightarrow -2I(2K - 1)(K + 1) = 2A(1 - 2K)$

$\Rightarrow (2K - 1)(2A) - 2I(2K - 1)(K + 1) = 0$

$\Rightarrow (2K - 1)[2A - 2I(K + 1)] = 0$

$\Rightarrow K = \frac{1}{2}$

Q5 (4)

$\operatorname{adj}(2A) = 2^2 \operatorname{adj} A$

$\Rightarrow \operatorname{adj}(\operatorname{adj}(2A)) = \operatorname{adj}(4 \operatorname{adj} A) = 16 \operatorname{adj}(\operatorname{adj} A)$

$= 16 \mid A l A$

$\Rightarrow \operatorname{adj}(32 \mid A l A) = (32|A|)^2 \operatorname{adj} A$

$12(32|A|)^2 \operatorname{ladj} A \mid = 2^3(32|A|)^6 \operatorname{ladj} A l$

$2^3 \cdot 2^{30}|A|^6 \cdot |A|^2 = 2^{41}$

$|A|^8 = 2^8 \Rightarrow |A| = \pm 2$

$|A|^2 = |A|^2 = 4$

Q6 (1)

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$A^n = \begin{bmatrix} 1 & 0 & 0 \\ n - 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$A^{2025} - A^{2020} = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^{6} - A = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q7 (2)

$$AA^{T} = \begin{pmatrix} \frac{1}{5} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$$

$$AA^{T} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Q^{2} = A^{T}BAA^{T}BA = A^{T}BIBA$$

$$\Rightarrow Q^{2} = A^{T}B^{2}\,A$$

$$Q^{3} = A^{T}B^{2}\,A\,A^{T}BA \Rightarrow Q^{3} = A^{T}B^{3}\,A$$

Similarly : $Q^{2021} = A^{T}B^{2021}\,A \ldots\ldots (1)$

Now $B^{2} = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix}$

$$B^{3} = \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} \Rightarrow B^{3} = \begin{pmatrix} 1 & 0 \\ 3i & 1 \end{pmatrix}$$

Similarly $B^{2021} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

$$\therefore AQ^{2021}\,A^{T} = AA^{T}B^{2021}AA^{T} = IB^{2021}I$$

$$\Rightarrow AQ^{2021}\,A^{T} = B^{2021} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$$

$$\therefore \left(AQ^{2021}\,A^{T}\right)^{-1} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$$